Zeros and Ones | Sahara

Health-a-thon 2026

About the Team / Organisation

A. **Team Zeros and Ones.** Shaik Abdul Basith (technical lead: website and the AI that writes down the conversation) and Shaik Muhammad Awaiz (Android app and server) built Sahara. Shirin Ayub (medical student) designed how patients log symptoms. Rituparna P, a lawyer, checked our form against India's advance-directive ("living will") format.

B. **Doctor partners.** Dr Sharada Vinod Kutty and Dr Sujay Halkur Shankar are pulmonologists (lung specialists) who treat many lung-cancer patients; Dr Sujay also works in palliative care (care that eases pain and distress in a serious illness). They chose this problem from their own clinics, wrote the clinical specification (the medical rules the app follows), set the order of the conversation checklist, and tested the app with us, most recently in the 3 Nov team meeting.

Problem Statement (use case specific)

Goals-of-care wishes are lost when they matter most. A goals-of-care conversation is when the doctor asks a seriously ill patient and family what treatment they want or refuse near the end of life: CPR (restarting the heart), a ventilator, ICU, dialysis, where to die. It is rarely written on the legal advance-directive form. Example: a 62-year-old woman with advanced breast cancer tells her doctor her wishes; two months later she is unconscious in the emergency department, her children disagree about CPR, and nobody can find what she said.

Evidence: about 15.6 lakh new cancer cases in India in 2024 (GLOBOCAN); 98% of patients with metastatic (spread) cancer had unmet palliative-care needs (Patil et al., 2024); 88% of surveyed urban patients wanted to choose their own end-of-life treatment (IndiaSpend, 2019); only 11.5% of doctors in one hospital were trained in advance directives (Jacob et al., 2026).

Users and setting: patients in palliative care and their families, at home; their palliative, oncology or lung doctor, in the clinic; and the emergency doctor, who has minutes to decide and needs the patient's wishes at once.

AI Solution Description

Sahara is a palliative-care app (Android and website). In the clinic the doctor leads the goals-of-care conversation; Sahara listens and writes. After the patient ticks "I agree", the doctor taps Record. Google's Gemini AI transcribes the talk (turns speech into text), labels who said what, and places each answer on the legal advance-directive form (word for word, as our lawyer checked). A checklist in the doctors' order ticks itself and, at the end, lists any topic not yet discussed. The doctor corrects it; doctor and patient or family sign on the phone, and the form locks: any later edit cancels the signatures. The signed wishes open from the patient's emergency QR code (a barcode any phone camera can scan). The AI only writes things down: the doctor makes every clinical decision.

Technology and Tools Used

Android (Kotlin, Jetpack Compose); website (Next.js, Vercel). **Supabase** is the server: sign-in, a Postgres database with row-level security (each family sees only its own records), and server functions that keep the AI keys off the phone. **Google Gemini** (backup models when busy) transcribes, fills the form and answers "Ask your documents" with the line it used. With no internet, **OpenAI Whisper** transcribes on the phone itself; if that fails, the doctor types. Google ML Kit: 5 languages and read-aloud on the phone.

Key Features of the solution

- **Goals-of-care conversation:** consent, record, transcribe, fill the legal form, sign and lock.
- **Emergency QR code and SOS:** wishes, contacts and medicines in one scan.
- **How I feel:** symptoms, a 0–10 distress scale (how upset the patient feels) and a pain body map; charts for the doctor.
- **Medicines and reports:** a tick for every dose, reminders, prescription photo; ask a question about a hospital report.
- **Help nearby and family:** palliative centres, ambulances, support groups; invited family share the record.

Build Sprint Progress

- **Weeks 1–2 (9–22 Oct):** shortlisted 9 Oct; native Android app, on Google Play closed testing from 14 Oct; map of nearby palliative centres; medicine organiser with prescription photo; symptom tracker.
- **Week 3 (23–29 Oct):** pain body map; hospital survey live 27 Oct; 28 Oct team meeting (9 decisions); “Ask your documents”; Dr Sharada’s clinical specification on 29 Oct.
- **Week 4 (30 Oct–5 Nov):** advance directive (living will) and emergency QR code; Telugu, Urdu, Kannada; legal review 31 Oct; Dr Sujay tested the whole app 3 Nov; goals-of-care recording with Gemini onto the legal form; signed, locked PDF.
- **How it changed:** in Round 1 the AI drafted a “Care Note”; our doctors asked that the doctor lead and the AI only write it down, on the legal form.

User Validation and AI Evaluation

- **Automatic tests:** 877 Android and 594/594 website tests pass (they re-check every feature after each change).
- **Goals of care:** 8 made-up conversations (6 languages, a change of mind, a family disagreement): every answer on the right line of the form, 8 of 8.
- **Ask your documents:** 4 of 4 questions on a sample hospital report answered with the right source line.
- **Dr Sujay, 3 Nov meeting:** asked for a guide on what to talk about next → checklist in the doctors’ order (consent, patient, family, CPR, dialysis…) with a “not yet discussed” prompt; 4 doses a day → 4 ticks. Team chat: tubes-and-drains tracker, darker text.
- **Rituparna P, lawyer, 31 Oct:** use the form’s exact legal wording → copied word for word.
- **Not tested yet:** no real-patient pilot; Gemini not yet measured on real Hindi or Telugu recordings; on-phone speech is weak for Telugu and Kannada, so typing is offered.

Safety and Responsible AI

- **Doctor in charge:** the doctor leads, corrects and signs every form; the AI never diagnoses or suggests treatment, and lists missing topics instead of guessing.
- **Consent and privacy:** recording starts only after the patient ticks “I agree”; Gemini is used only with that consent. Family members join by invite and see only their patient.
- **Tamper-evident forms:** signed forms are sealed with a SHA-256 fingerprint, so any change cancels the signatures; earlier versions stay visible.
- **Inclusion:** five languages, read aloud, large text, thumb signature; typing when speech fails. Demos use made-up patients only.

Deployment and Sustainability Plan

- **Pilot:** 60–90 days in one palliative-care clinic with our doctors; patients consent as India’s data-protection law (DPDP Act 2023) requires. We count talks signed and how fast wishes are found.
- **Integration:** works on its own, no hospital software needed; a FHIR export (the standard format hospital systems read) is ready for later.
- **Cost (team estimate):** servers about ₹3,900 a month (Supabase, Vercel); Gemini about ₹2 per 30-minute talk; about ₹1,600 per clinic a month if 10 clinics share it. Free for patients; clinics or NGOs pay ₹1,500–2,000 a month.

Supporting Links

- MVP: sahara.wedevit.in · Android app on Google Play
- Video Demo Walkthrough: link to be added
- Other Supporting Links: code (private, access on request)